

TITLE OF PROJECT

“ALFA: Adaptive Learning Framework for Agents — an agent that learns from its own outcomes, instantiated on a financial corpus”

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JIIT logo goes here — save the institute logo
as **docs/synopsis/logo.png** and re-run
./build.sh. Everything else on this page is
already positioned for it.

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1. Motivation Behind the Project

An agent built on a foundation model is frozen on the day it ships. The accepted way to improve one is to retrain or replace the model underneath it — expensive, and unattributable: when the answers change, nobody can say whether the product improved or the vendor swapped the weights.

ALFA is a framework, not one application. It is an agent layer that attaches to a body of data, answers questions over it, and **improves from its own interaction, outcomes and feedback while the foundation model's weights stay frozen.** The domain enters at one seam only — the documents you add, plus one numeric head per quantity that domain measures (Figure 1) — so a second domain is a new data pack, not a new agent.

It is **currently instantiated on a financial corpus** we collected ourselves (100.8M tokens, 4,674 documents, 101 instruments), because finance is the harshest cheap test of both promises: answers contain numbers that are either right or wrong, and a dated public record exists to check them. Nothing in the loop is financial. The research question is narrow and testable: *can an agent improve its decisions and guidance from interaction, outcomes and feedback without continuously retraining its underlying foundation model?*

2. Type of Project

Research cum Development Project

- **Research** → adaptation from feedback with frozen weights, measured against non-learning controls, over a transformer stack written from first principles (hand-derived autograd, no deep-learning framework)
- **Development** → the framework packages — agent core, retrieval, registry, database — an offline serving layer and a dashboard over it
- **Domain-agnostic:** one data seam; finance is the reference pack, not the product
- **Constraint:** no paid API and no API key anywhere — the model is built, not called

3. Critical Analysis of Research Papers & Gaps

Paper	Key Idea	Gap	One-Line Summary
LSTM / GRU forecasting (2020)	Sequence model over a numeric history	Scored against weak baselines, not persistence	Useful head, easy to over-credit
FinBERT / domain sentiment (2021)	Domain text classification	Score never joined to the decision	Strong NLP, unconnected to advice
RAG / Fusion-in-Decoder (2020–22)	Generate from retrieved evidence	Improves fluency, not truth of figures	Grounded in form, not in fact
RLHF and adapter tuning (2021–22)	Learn from judged outcomes	Both change weights by gradient: cost, drift, no attribution	Effective, not attributable
LLM agent frameworks (2023–25)	Tools and memory around a hosted model	Nothing measures whether the agent improved	Plumbing, not learning

Gap Identified:

No framework puts these four together: **(i)** one data seam, so the same agent serves any corpus, **(ii)** measurable improvement from its own logged outcomes with the foundation model's weights frozen, **(iii)** every figure verified against the evidence that produced it, and a named refusal when it is not, and **(iv)** every claim reported against the strongest cheap baseline, negative results included.

4. Overall Design of Project

Architecture Overview

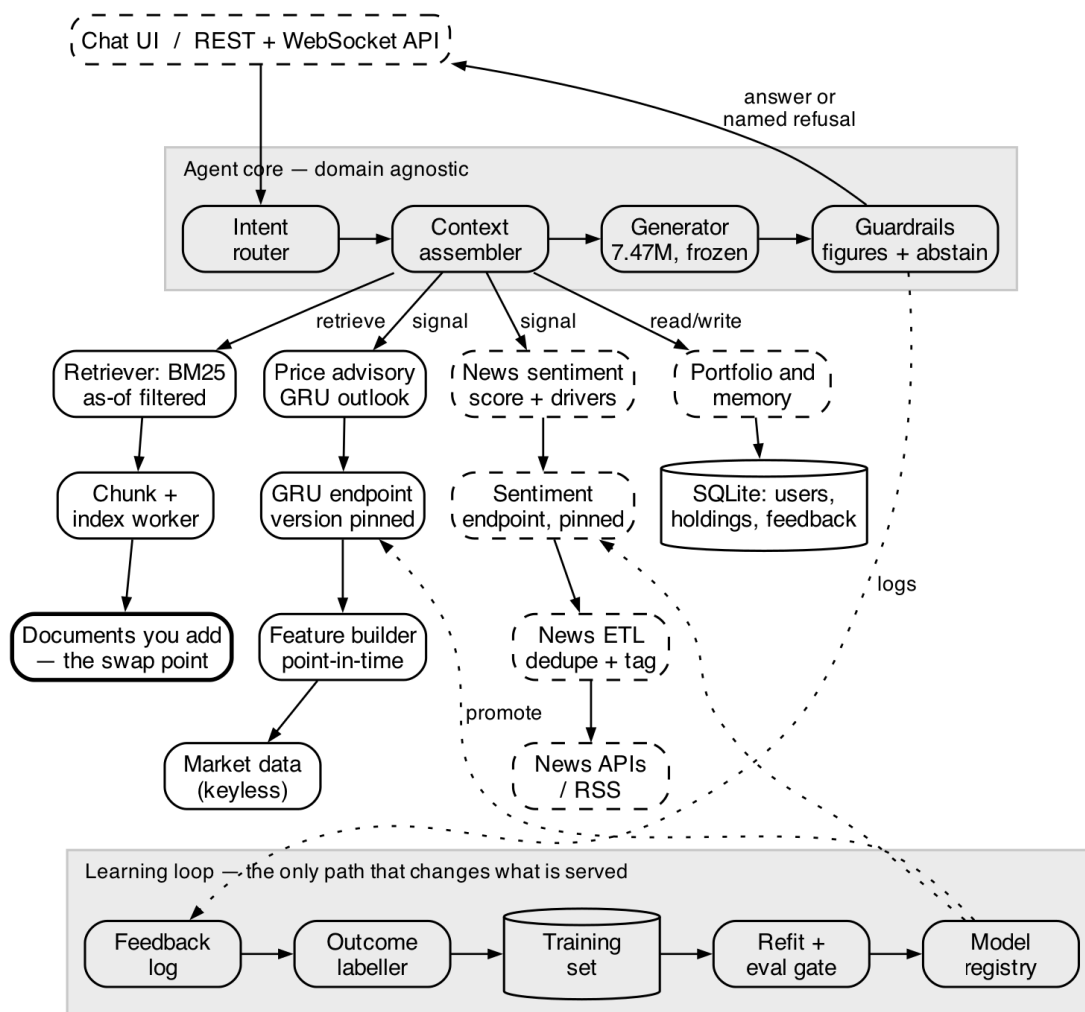


Figure 1 (Project Architecture)

The agent core, the guardrails and the learning loop know nothing about finance: a domain reaches them through the bold box — *the documents you add* — and one numeric head per quantity it measures. **Dashed boxes are specified and queued** (news and sentiment, portfolio and memory, the eval gate, the REST/WebSocket layer, the dashboard); solid boxes run offline in the repository today. The loop is the only path that changes what is served, and what it changes is a threshold, a calibrator and a registry pointer — never a weight.

Key Components (Section 4, continued):

- **Agent core (domain agnostic):** intent router → context assembler → generator → guardrails
- **Data pack (finance today):** feature builder, GRU volatility head, BM25 index over 44,811 chunks
- **Models:** pure-NumPy transformer encoder–decoder, 7,472,192 parameters; GRU head, 202,755 — version-pinned in a registry behind an eval gate
- **Learning loop:** SQLite feedback log → outcome labeller → refit of the cut, the calibrator and memory
- **Serving and UI:** FastAPI REST + WebSocket; Next.js and TypeScript dashboard (in progress)
- **Data layer:** keyless public endpoints, official APIs and RSS, every URL and licence in a manifest

5. Features Built & Languages Used

Framework Features (domain agnostic)

- Conversational question answering over the attached corpus, offline, in about 1.3 seconds
- Intent routing with an out-of-domain gate that refuses questions the data cannot answer
- Guardrails: one unsupported number turns the answer into a named refusal, and an abstention cut over calibrated confidence decides when to stay quiet
- Point-in-time discipline: no record later than the as-of stamp is ever read
- Experience store and per-user memory: one row per interaction — state, action, outcome, reward, policy version
- A version-pinned registry with an eval gate; swapping the data pack changes domain, leaving the core, the guards and the loop untouched
- 407 automated tests covering every gate above

Finance Pack (the current instantiation)

- A 100.8M-token corpus collected politely: robots.txt honoured, one request at a time, licence and URL per file
- Technical indicators computed by arithmetic, so the facts quoted are exact
- Week-ahead volatility outlook from a GRU head, served only when it beats arithmetic by more than its own noise; no buy or sell call — direction measured unpredictable
- News sentiment, portfolio and holdings memory, and a streaming dashboard (in progress)

Languages & Technologies

- **Frontend:** React, Next.js, TypeScript
- **Backend:** Python (FastAPI)
- **ML/NLP:** NumPy only — autograd, attention, GRU, tokeniser and optimiser written in the project; no TensorFlow, PyTorch or HuggingFace in the model path
- **Database:** SQLite · **Testing:** pytest

6. Proposed Methodology

Pipeline Flow:

1. Attach a data pack: a keyless public corpus, URL and licence recorded per file (today 100.8M tokens of prices, filings and news)
2. Pretrain the encoder–decoder on that corpus as a masked language model
3. Build question–answer pairs over point-in-time snapshots; split by date, subject, wording
4. Fine-tune with Fusion-in-Decoder cross-attention, so it phrases evidence rather than recalling it
5. Add one numeric head per quantity the domain measures; keep it only if it beats the cheapest baseline
6. Route to an intent, refuse if out of domain, rewrite into the trained phrasing
7. Assemble context from memory, passages and point-in-time state; verify every figure against it
8. Judge the turn, log one experience record, refit the cut and calibrator — weights frozen
9. Serve over REST and WebSocket; register each artifact behind the eval gate (in progress)

Steps 6–9 never mention the domain: a second domain repeats steps 1–5 only.

7. Algorithm / Description of Work

What is allowed to learn, and what is not

Four things stay separate, because the experiment collapses if any two are entangled: **knowledge** (frozen encoder plus retrieved documents), **reasoning** (code, not weights), **policy** (whether and when to answer — the only thing that learns) and **memory** (per user, never shared). The encoder's parameters never appear in an optimiser — a testable statement, and it has a test.

Adaptation Algorithm (the research contribution)

The agent is given a promise — *answer only where you are at least p likely to be right* — and solves for the threshold that delivers it from its own logged outcomes:

$$c^* = \min \{ c : \text{precision}(\text{confidence} \geq c) \geq p \} \text{ over the feedback log, with } \hat{p} = \sigma(a \cdot \logit(\text{confidence}) + b)$$

Only c^* , a and b are fitted; no gradient is computed. That is what makes an improvement attributable to feedback rather than to retraining.

Grounded Generator (frozen, domain agnostic)

7,472,192 parameters; masked-language pretraining, then supervised fine-tuning. The decoder cross-attends to the evidence, and one unsupported number replaces the answer with a refusal.

Numeric Head (finance pack)

A 128-bar window with six channels, standardised inside the window so nothing after the as-of date can enter it; the output is the next five days' volatility in three buckets. 202,755 parameters over 143,546 windows from 101 instruments, served only when its recorded accuracy beats persistence by more than one standard error. Five-day *direction* measured unpredictable, so there is no BUY / HOLD / SELL output.

8. Division of Work Among Students

Member	Role	Contribution
Aditya Pachouri	Frontend Developer	UI design, dashboard, charts, API integration
Aditya Pratap Singh	Backend Developer	FastAPI REST and WebSocket, data collection, integration
Rishabh Kapur	ML Engineer	Autograd and transformer stack, numeric head, adaptation loop

9. Results

What was measured	Result
Feedback-driven adaptation, weights frozen	Keeps its stated 60% promise to within 8.5 points , vs 16.7 for a fixed threshold and 29.2 for answering everything
Intent routing, out-of-domain gate	94.3% correct intent; refuses 12/12 out-of-domain questions at 88.5% coverage
Rewriting into the trained phrasing	Exact match 33.5% → 73.5% on held-out wordings, frozen weights
Does the generator read its evidence?	Blanking the evidence costs +1.04 loss, swapping it +1.95 ; the first version cost +0.018 and was ignoring it
Unsupported figures in answers	1.7% on trained wording, 4.7% on reworded
Numeric head, 28,676 held-out windows	Volatility: GRU 49.1% vs persistence 47.8%, majority 39.6%. Direction 35.7% vs 36.0% — unpredictable , never advised on
Retrieval recall@5, 44,811 chunks	BM25 15.5% , hybrid 12.0%, dense 1.5% — the vector arm was dropped
Confidence calibration	Calibration error 0.683 → 0.121 ; ranking AUC 0.740 ± 0.039
System	407 tests passing; a full answer in ~1.3 s on a laptop CPU, offline, no API key

Observation:

- The thesis holds in a narrow, defensible form: the agent **learns when to speak, not what to say** — eight samples give only 1.77 distinct answers, so the wording has nothing to improve.
- The larger gain came from the input, not the loop: rewriting a question into wording the decoder knows is worth 40 points against the loop's ~4.

10. Conclusion

ALFA trains its own foundation model and then improves from its logged outcomes without ever retraining it. Every figure in an answer is checkable against the evidence that produced it, refusals name their reason, adaptation is attributable because the weights are frozen, and a new domain attaches at one seam — the corpus you add. The limitation is measured, not assumed: an unfamiliar sentence *shape* is handled only 68% of the time. Finance is the first pack, not the product — the end product is this framework with the news, portfolio, registry and serving stages of Figure 1 completed.